MTH632

Final term solution file:

Sabin Saeed

Q1: Prove that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ satisfies Laplace's equation.

Sol: Given $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 6x \quad ; \quad \frac{\partial u}{\partial y} = -6xy - 6y$$

$$\frac{\partial^2 u}{\partial x^2} = 6x + 6 \qquad \frac{\partial^2 u}{\partial y^2} = -6x - 6$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 6 - 6x - 6 = 0$$

$\therefore$ u satisfies Laplace's equation.

Q2: Show that

(a) $\text{Log}(1+i)^2 = 2\text{Log}(1+i)$

(b) $\text{Log}(-1+i)^2 \neq 2\,\text{Log}(-1+i)$

Sol:

(a) $(1+i) = \sqrt{2}\left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right) = \sqrt{2}\left(\text{Cos}\frac{\pi}{4} + i\,\text{Sin}\frac{\pi}{4}\right)$

$$= \sqrt{2}\, e^{i\pi/4}$$

$$(1+i)^2 = \left(\sqrt{2}e^{i\pi/4}\right)^2 = 2e^{i\pi/2}$$

therefore $\text{Arg}(1+i) = \frac{\pi}{4}$ and $\text{Arg}(1+i)^2 = \frac{\pi}{2}$

so that

$$\text{Log}(1+i)^2 = \ln|(1+i)^2| + i\,\text{Arg}(1+i)^2$$

$$= \ln|2e^{i\pi/2}| + i\frac{\pi}{2}$$

$$= \ln\left|2\left(\cos\frac{\pi}{2} + i\sin\frac{2\pi}{2}\right)\right| + i\frac{\pi}{2}$$

$$= \ln|2| + i\frac{\pi}{2}$$

"x" and "÷" by 2

$$= 2\left(\frac{1}{2}\ln|2| + i\frac{\pi}{4}\right) = 2\left(\ln\sqrt{2} + i\frac{\pi}{4}\right) = 2\cdot\text{Log}(1+i)$$

(b) $-1+i = \sqrt{2}\left(\frac{-1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right) = \sqrt{2}e^{i3\pi/4}$

$$(-1+i)^2 = -2\cdot i = 2e^{-i\pi/2}$$

therefor

$$\text{Arg}(-1+i) = \frac{3\pi}{4} \quad ; \quad \text{Arg}(-1+i)^2 = \frac{\pi}{2}$$

so that

$$\text{Log}(-1+i)^2 = \ln|(-1+i)^2| + i\text{Arg}(-1+i)^2$$

$$= \ln|2e^{-i\pi/2}| + i\frac{\pi}{2}$$

$$= \ln|2| + i\frac{\pi}{2} \quad \text{---(i)}$$

$$2\,\text{Log}(-1+i) = \ln 2 + i\frac{3\pi}{2} \quad \text{---(ii)}$$

from (i) and (ii)

$$\Rightarrow \quad \text{Log}(-1+i)^2 \neq 2\,\text{Log}(-1+i)$$

Q3: $\frac{d}{dz}\arcsin z = \frac{1}{\sqrt{1-z^2}}$

Sol: Let $w = \arcsin z$

$$z = \sin w$$

Take derivative of b/s

$$\frac{d(z)}{dz} = \frac{d(\sin w)}{dz}$$

$$1 = \frac{d}{dw}\sin w \cdot \frac{dw}{dz}$$

$$1 = \cos w \cdot \frac{dw}{dz}$$

$$\frac{1}{\cos w} = \frac{dw}{dz}$$

$$\frac{dw}{dz} = \frac{1}{\sqrt{1-\sin^2 w}}$$

$$\frac{dw}{dz} = \frac{1}{\sqrt{1-z^2}}$$

$$\Rightarrow \frac{d}{dz}\,\text{arc}\sin z = \frac{1}{\sqrt{1-z^2}} \quad \text{(proved)}$$

Q4: Prove that $\int_0^{\pi+2i} \cos\left(\frac{z}{2}\right) dz = e + 1/e$

Sol: $L.H.S = \int_0^{\pi+2i} \cos\left(\frac{z}{2}\right) dz$

$$= \left| \frac{\sin(z/2)}{1/2} \right|_0^{\pi+2i} = 2\left| \sin\left(\frac{z}{2}\right) \right|_0^{\pi+2i}$$

$$= 2\left[\sin\left(\frac{\pi+2i}{2}\right) - \sin(0) \right]$$

$$= 2\left[\sin\left(\frac{\pi}{2}+i\right) \right]$$

$$= 2\left[\frac{e^{i(\frac{\pi}{2}+i)} - e^{-i(\frac{\pi}{2}+i)}}{2i} \right]$$

$$= \frac{1}{i}\left[e^{(i\pi/2+i^2)} - e^{(-i\pi/2-i^2)} \right]$$

$$= \frac{1}{i}\left[e^{i\pi/2-1} - e^{-i\pi/2+1} \right]$$

$$= \frac{1}{i}\left[e^{i\pi/2}\cdot e^{-1} - e^{-i\pi/2}\cdot e^{1} \right]$$

$$= \frac{1}{i}\left[\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)e^{-1} - \left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)e \right]$$

$$= \frac{1}{i}\left[(0+i)e^{-1} - (0-i)e^{1}\right]$$

$$= \frac{1}{i}\left(i\frac{1}{e} + ie\right)$$

$$= \frac{i}{i}\left(\frac{1}{e} + e\right) = R.H.S$$

proved.

Q5: $z_n = \frac{1}{n^3} - i$ solve it for convergence also find value of ε.

Sol: $z_n = \frac{1}{n^3} - i$

$$\lim_{n\to\infty} z_n = \lim_{n\to\infty}\left(\frac{1}{n^3} - i\right)$$

$$= \lim_{n\to\infty}\frac{1}{n^3} - \lim_{n\to\infty}(i)$$

$$= \frac{1}{\infty} - i = -i \angle 0 \text{ (converges)}$$

Q6: Evaluate $\bar{z}_1 - \bar{z}_2$ when $z_1 = 5+i$ and $z_2 = -8+3i$

Sol: $z_1 = 5+i \quad ; \quad z_2 = -8+3i$

$\bar{z}_1 = 5-i \quad ; \quad \bar{z}_2 = -8-3i$

$\bar{z}_1 - \bar{z}_2 = (5-i) - (-8-3i)$

$= 5 - i + 8 + 3i$

$= 13 + 2i$

$= 2(6+i)$

Q7: If $\int_C f(z)\,dz$ & $f(z) = \frac{z+2}{z}$

and $z = 2e^{i\theta}$

Sol: $\int_C f(z)\,dz$

$= \int_0^{\pi} \left(\frac{z+2}{z}\right) dz$ $\quad \because z = 2e^{i\theta}$, $dz = 2ie^{i\theta}\,d\theta$

$= \int_0^{\pi} \left(1 + \frac{2}{z}\right) dz = \int_0^{\pi} \left(1 + \frac{2}{2e^{i\theta}}\right)(i2e^{i\theta})\,d\theta$

$= \int_0^{\pi} \left(1 + \frac{1}{e^{i\theta}}\right)(2ie^{i\theta})\,d\theta$

$= 2i \int_0^{\pi} \frac{e^{i\theta}+1}{e^{i\theta}} \cdot e^{i\theta}\,d\theta$

$= 2i \left[\int_0^{\pi} e^{i\theta}\,d\theta + \int_0^{\pi} d\theta\right]$

$= 2i \left[\frac{e^{i\theta}}{i}\right|_0^{\pi} + 2i\,\theta\Big|_0^{\pi}$

$= 2(e^{i\pi} - e^{0}) + 2i\pi$ $\quad \because e^{i\pi} = \cos\pi + i\sin\pi$

$= 2e^{i\pi} - 2 + 2i\pi$ $\quad = -1 + 0$

$= -2 - 2 + 2i\pi$

$= -4 + 2i\pi$

Q8: If $x(t) = +a\sin t\,\hat{e}_1 + a\cos t\,\hat{e}_2 + bt\,\hat{e}_3$

then solve $\frac{x'(t)}{|x'(t)|}$

Sol: Given $x(t) = a\cos t\,\hat{e}_1 + a\sin t\,\hat{e}_2 + bt\hat{e}_3$

$$x'(t) = -a\sin t\,\hat{e}_1 + a\cos t\,\hat{e}_2 + b\hat{e}_3$$

$$|x'(t)| = \sqrt{a^2\sin^2(t) + a^2\cos^2(t) + b^2}$$

$$= \sqrt{a^2(\sin^2 t + \cos^2 t) + b^2}$$

$$= \sqrt{a^2 + b^2}$$

we know that

$$\hat{t} = x = \frac{x'(t)}{|x'(t)|} = \frac{-a\sin(t)\hat{e}_1 + a\cos(t)\hat{e}_2 + b\hat{e}_3}{\sqrt{a^2+b^2}}$$

Q9: If $p = (1,1,0)$ and $v = (1,0,-3)$,

$f = xy^2z$

Then find $V_p[f]$.

Sol: 1st method:

$$(P+tv) = (1,1,0) + t(1,0,-3)$$

$$= (1+t, 1, -3t)$$

$$f(P+tv) = f(1+t, 1, -3t)$$

$$= (1+t)^2(1)(-3t)$$

$$= -3t^3 - 6t^2 - 3t$$

$$\frac{d}{dt}(f(P+tv)) = \frac{d}{dt}(-3t^3 - 6t^2 - 3t)$$

$$= -9t^2 - 12t - 3$$

Applying definition of $V_p[f]$

$$V_p[f] = \frac{d}{dt}(f(P+tv))\Big|_{t=0}$$

$$= -9t^2 - 12t - 3\Big|_{t=0}$$

$$= -3$$

So -3 is directional derivative.

2nd method:

$$f = x^2yz$$

$$\frac{\partial f}{\partial x} = 2xyz \;,\quad \frac{\partial f}{\partial y} = x^2z \;,\quad \frac{\partial f}{\partial z} = x^2y$$

at point $P=(1,1,0)$

$$\frac{\partial f}{\partial x} = 0 \;,\quad \frac{\partial f}{\partial y} = 0 \;,\quad \frac{\partial f}{\partial z} = 1$$

Here

$$V_p[f] = \sum v_i \frac{\partial f}{\partial x_i}(P)$$

$$= 1(0) + 0(0) \mp 3(1)$$

$$= -3$$

-3 is directional derivative.

Q10: Find singularity if $f(z) = \frac{1}{z^2+2z+2}$

Sol: Here $z^2+2z+2=0$

applying quadratic formula

$$z = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-(2) \pm \sqrt{(2)^2-4(1)(2)}}{2(1)}$$

$$= \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2} = -1 \pm i$$

$z = -1 \pm i$ is singular point.

Q11: Prove that $\frac{d}{dz}(\sin z) = \cos(z)$

Sol: $L \cdot H \cdot S = \frac{d}{dz}(\sin z)$

$$= \frac{d}{dz} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$$

$$= \sum_{n=0}^{\infty} \frac{d}{dz} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)}{(2n+1)2n!} \cdot z^{2n}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n!} \cdot z^{2n}$$

$$= \cos z = R \cdot H \cdot S$$

Q#12: Show that $\sum_{n=1}^{\infty} \bar{z}_n = \bar{S}$

Sol: We know that

$$\bar{z}_n = x_n - iy_n$$

and $S = X + iY$; $\bar{S} = X - iY$

$$\sum_{n=1}^{\infty} x_n = X \quad \text{and} \quad \sum_{n=1}^{\infty} y_n = Y$$

$$\sum_{n=1}^{\infty} \bar{z}_n = \sum_{n=1}^{\infty} (x_n - iy_n)$$

$$= \sum_{n=1}^{\infty} x_n - i \sum_{n=1}^{\infty} y_n$$

$$= X - iY$$

$$\sum_{n=0}^{\infty} \bar{z}_n = \bar{S} \quad \text{proved}$$

Q 13: Show that $(-1)^{1/\pi} = \exp(1+2n)i$

Sol: $L \cdot H \cdot S = (-1)^{1/\pi}$

$= \exp\left[\frac{1}{\pi}\log(-1)\right]$

$= \exp\left[\frac{1}{\pi}\log(1)\right]$

$= \exp\left[\frac{1}{\pi}(\ln|1| + i[\text{Arg}(1) + 2n\pi])\right]$

$= \exp\left[\frac{1}{\pi}(0 + (\pi + 2n\pi)i)\right]$

$= \exp\left[\frac{1}{\pi} \cdot \pi(1+2n)i\right]$

$= \exp(1+2n)i$ R·H·S

Q 14: Show that $\overline{\cos(iz)} = \cos(\bar{i}\bar{z})$

Sol: L·H·S $\overline{\cos(iz)}$

$= \overline{\cos(-y+ix)}$

$= \cos y \cosh x - \sin y \sinh x$ — (a)

$z = x + iy$

$iz = ix - y$

$\bar{i}\bar{z} = -ix - y$

R·H·S $= \cos(\bar{i}\bar{z})$

$= \cos(-ix - y)$

$= \cos(-(y+ix))$ $\because \cos(-x) = \cos x$

$= \cos(y+ix)$

$= \cos y \cosh x - \sin y \sinh x$ — (b)

From (a) and (b)

L·H·S = R·H·S

Q 15: prove that $d(fg) = gfd + f\,dg$

Sol: $d(fg) = \sum \frac{\partial(fg)}{\partial x_i} dx_i$

$= \sum\left(\frac{\partial f}{\partial x_i} \cdot g + \frac{\partial g}{\partial x_i} \cdot f\right) dx_i$

$$= \sum \frac{\partial f}{\partial x_i} \cdot g \, dx_i + \sum \frac{\partial g}{\partial x_i} \cdot f \, dx_i$$

$$= g \sum \frac{\partial f}{\partial x_i} dx_i + f \sum \frac{\partial g}{\partial x_i} dx_i$$

$$= g\,df + f\,dg \quad \text{(proved)}$$

Q 16: Find df and $df(v_p)$ if

$$f = (x^2-1)y + (y^2-1)x$$

Q17: Show that the function $f(z)=\bar{z}$ nowhere differentiable

Sol: Given $u+iv = x-iy$

$u = x$; $v = -y$

$u_x = 1$; $v_x = 0$

$u_y = 0$; $v_y = -1$ $\therefore$ $u_x \neq v_y$

$\therefore$ C-R eq. are not satisfied

$f(z)=\bar{z}$ nowhere differentiable.

Q18: Test the analyticity of function $w=\sin z$

Sol: $w = f(z) = \sin z$

$u+iv = \sin(x+iy)$

$(u+iv) = \sin x\cosh y + i\cos x\sinh y$

Now

$u = \sin x\cosh y$; $v = \cos x\sinh y$

$u_x = \cos x\cosh y$; $v_x = -\sin x\sinh y$

$u_y = +\sin x\sinh y$; $v_y = \cos x\cosh y$

$u_x = u_y$ & $-v_x = u_y$

C-R equation satisfied.

$\therefore$ function is analytic.

Q19: Show that $f(z)=e^z$ is analytic

Sol: $f(z)=e^z$

$u+iv = e^{x+iy}$

$u+iv = e^x \cdot e^{iy} = e^x(\cos y + i\sin y) = e^x\cos y + ie^x\sin y$

$u = e^x\cos y$; $v = e^x\sin y$

$u_x = e^x\cos y$; $v_x = e^x\sin y$

$u_y = -e^x\sin y$; $v_y = e^x\cos y$

$u_x = v_x$ & $u_y = -v_x \Rightarrow$ function is analytic

Q 20: Give an example such that u and v are harmonic but u+iv is not analytic.

Sol: Consider the function $\omega = \bar{z} = x - iy$

$u = x$; $v = -y$

$u_x = 1$ $\quad$ $v_y = -1$

$u_{xx} = 0$ $\quad$ $v_{yy} = 0$

$u_y = 0$ $\quad$ $v_x = 0$

$u_{yy} = 0$ $\quad$ $v_{xx} = 0$

$u_x \neq v_x$

$\therefore$ f(z) is not analytic.

But $u_{xx} + u_{yy} = 0$ and $v_{xx} + v_{yy} = 0$ gives u and v are harmonic.

Q #21: Write $-3i^4$ in the form of a+ib.

Sol. Let $z = -3i^4$

$z = -3(i^2)^2$

$z = -3(-1)^2 = -3$

$z = -3 + 0i$

Q 22: Find antiderivative of $\int_i^{i/2} e^{\pi z}\, dz$.

Sol: $\int_i^{i/2} e^{\pi z}\, dz$

$= \frac{e^{\pi z}}{\pi}\Big|_i^{i/2} = \frac{1}{\pi}\left[e^{i\pi/2} - e^{i\pi}\right]$

$\because e^{i\pi} = \cos\pi + i\sin\pi$

$= -1 + 0 = -1$

$\because e^{i\pi/2} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$

$= \frac{1}{\pi}\left[i - (-1)\right] = \frac{i+1}{\pi}$

$= 0 + i = i$

Q 23: Prove that $u = \frac{1}{2}\log(x^2+y^2)$ is harmonic and find its conjugate harmonic.

Sol: Given $u = \frac{1}{2}\log(x^2+y^2)$

Differentiate partially w.r.t. "x" and "y"

$$\frac{\partial u}{\partial x} = \frac{x}{x^2+y^2} \quad ; \quad \frac{\partial u}{\partial y} = \frac{y}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{y^2-x^2}{(x^2+y)^2} \quad ; \quad \frac{\partial^2 u}{\partial y^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{y^2-x^2+x^2-y^2}{(x^2+y^2)^2} = 0$$

$\Rightarrow u$ is harmonic

Let $v(x,y)$ be the conjugate harmonic. Then $w = u+iv$ is harmonic

By C-R equations, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

we have

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$

$$dv = -\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$$

$$dv = \frac{-y}{x^2+y^2}dx$$

Integrating

$$v = \tan^{-1}(y/x) + c.$$